PRODUCT REQUIREMENTS DOCUMENT

AI for the Indian Investor

Chart Pattern Intelligence — NSE Universe

Turning Market Data into Actionable Investment Decisions

Product	AI for the Indian Investor
Module	Chart Pattern Intelligence
Version	1.0 — Initial Release
Platform	ET Markets (Web + Mobile)
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Status	Draft — For Review

ET Markets · Times Internet Limited

1. Executive Summary

The One-Line Vision

Build the AI intelligence layer on top of ET Markets' data to give India's 14 crore+ retail investors the same edge that professional fund managers and institutional desks have always enjoyed — starting with real-time chart pattern detection and plain-English signal explanations across the entire NSE universe.

India is in the midst of an unprecedented retail investing boom. Demat accounts have crossed 14 crore, SIP inflows consistently exceed ₹20,000 crore per month, and millions of first-generation investors are entering capital markets every year. Yet the tools available to these investors lag far behind. Most retail participants trade on tips from WhatsApp groups, YouTube influencers, or broker newsletters — not because they lack intelligence, but because genuinely useful analytical tooling is either inaccessible, unaffordable, or requires expertise they have not yet acquired.

ET Markets sits at the intersection of the world's largest financial data corpus on Indian equities and the largest readership of Indian financial news. This PRD defines the first module of the AI for the Indian Investor product suite: Chart Pattern Intelligence — a real-time technical analysis engine that detects high-probability patterns, explains them in plain English, and surfaces back-tested success rates so every investor can make faster, more confident, and better-informed decisions.

2. Problem Statement

2.1 The Retail Investor's Reality

India has more than 14 crore demat account holders, yet the vast majority of them operate in a persistent information asymmetry. The following problems are endemic and well-documented across the investor base:

Problem	Description
Tip-driven trading	Most retail investors rely on unverified tips from social media, broker calls, or news headlines — leading to reactive, emotion-driven entries and exits with no analytical basis.
Technical literacy gap	Reading charts — identifying RSI divergences, Fibonacci levels, or candlestick patterns — requires months of learning. Most retail investors never bridge this gap and miss high-probability setups daily.
Filing blindness	NSE and BSE publish thousands of exchange filings, corporate announcements, and regulatory disclosures every week. No individual investor has the bandwidth to monitor even a watchlist, let alone the full market.

Gut-feel MF management	Mutual fund investors frequently rebalance based on NAV panic rather than fundamentals, missing long-term compounding and making value-destroying decisions at market bottoms.
No personalized context	Generic market news cannot be personalised to an investor's existing holdings, risk profile, or investment horizon. The same article is read by a day trader and a 10-year SIP investor, yet both need entirely different takeaways.

2.2 The Opportunity for ET Markets

ET Markets is uniquely positioned to solve this problem. It possesses three strategic advantages that no pure-play fintech startup can replicate in the near term:

- Data Depth: Decades of NSE/BSE tick data, OHLCV history, corporate filings, and news corpus.
- Distribution: Tens of millions of monthly active users across the ET Markets web and app ecosystem.
- Trust: The Economic Times brand is the most trusted financial media entity in India.

The missing piece is intelligence — the layer that transforms raw data into personalised, actionable, and explainable signals. That is precisely what this product builds.

3. Solution Overview

3.1 Product Vision

"Give every Indian retail investor the analytical superpower of a full-time professional analyst — available 24/7, at zero marginal cost, in plain Hindi and English."

The AI for the Indian Investor product suite is an intelligence layer built on top of ET Markets' existing data infrastructure. It does not replace the news feed or the market data ticker — it adds a persistent analytical companion that watches the market so the investor does not have to.

3.2 Module Architecture (Phased Roadmap)

The product is designed as a modular intelligence suite. This PRD covers Module 1 in full detail. Subsequent modules are outlined for roadmap context.

Phase	Module	Timeline	Core Value

1	**Chart Pattern Intelligence**	Q2 2026	Real-time NSE-wide pattern detection with plain-English explanations and back-tested success rates
2	Smart Filing Monitor	Q3 2026	AI-parsed BSE/NSE filings summarised into one-line investor-grade alerts with sentiment tagging
3	MF Portfolio Analyser	Q3 2026	AI-driven overlap analysis, rebalancing suggestions, and goal-tracking for mutual fund investors
4	Personal Market Assistant	Q4 2026	Conversational AI that answers questions about holdings, news context, and portfolio decisions

4. Feature Specification — Chart Pattern Intelligence

4.1 Feature Overview

What This Feature Does

Chart Pattern Intelligence continuously scans every actively traded security on the NSE — all ~2,000 equities — using real-time OHLCV data and identifies high-probability technical patterns the moment they form or confirm. For each detected pattern, it generates a plain-English explanation, a visual annotation overlay, and a back-tested success rate specific to that pattern on that stock.

4.2 User Stories

The following user stories define the scope of this feature from the investor's perspective:

1. As a retail equity investor, I want to see which of my watchlist stocks are forming breakout patterns today, so that I can plan my entry before the move happens rather than chasing it.
2. As a new investor, I want each detected pattern explained in simple language — no jargon — so that I understand why the pattern matters and what it suggests about price direction.
3. As an active trader, I want to know the historical success rate of a specific pattern on a specific stock, so that I can size my position proportional to the probability of the trade working out.
4. As a long-term investor, I want weekly pattern summaries for my portfolio stocks, so that I can make informed add/trim decisions without monitoring charts every day.
5. As a user on mobile, I want push notifications when a high-probability pattern confirms on a stock I follow, so that I never miss a setup because I was away from my desk.

4.3 Patterns Detected — Scope of Coverage

The initial release will detect and classify the following pattern families. Patterns are organized by analytical category:

Category	Patterns (Phase 1)	Patterns (Phase 2 — Future)
Breakout Patterns	Cup and Handle, Ascending Triangle, Rectangle Breakout, 52-Week High Breakout	Volatility Contraction Pattern (VCP), Base Breakout
Reversal Patterns	Head and Shoulders (Top/Bottom), Double Top/Bottom, Triple Top/Bottom	Rounded Bottom, Adam & Eve Double Bottom
Support / Resistance	Dynamic Support/Resistance, Horizontal S/R levels, Pivot Points	Volume-Profile-based S/R, VWAP Levels
Candlestick Patterns	Doji, Hammer, Engulfing (Bull/Bear), Morning/Evening Star, Shooting Star	Three White Soldiers, Three Black Crows, Harami Cross
Divergences	RSI Divergence (Regular + Hidden), MACD Divergence	OBV Divergence, Stochastic Divergence
Momentum Signals	Golden Cross / Death Cross, EMA Crossovers (9/21, 50/200)	Supertrend Signal, ADX Trend Strength

4.4 The Explanation Engine — Plain-English Output

The core differentiator of this feature is not pattern detection (which existing screeners partially do) — it is explanation. Every detected pattern generates a structured output card containing the following elements:

Output Field	Description
Pattern Name	Human-readable label e.g. "Cup and Handle Breakout" with a tooltip-style glossary definition.
Plain-English Signal	2–3 sentence explanation in investor-grade plain English: what the pattern means, why it matters, and what it typically suggests about near-term price action. Available in English and Hindi.
Back-tested Success Rate	Historical win rate for this exact pattern on this specific stock, calculated over the last 5 years of data. E.g. "This pattern has resolved bullishly 71% of the time on Reliance Industries over 23 historical occurrences."
Suggested Action Zone	AI-calculated entry price range, stop-loss level, and target price based on pattern geometry and ATR-adjusted volatility. Framed as a suggestion, not a buy/sell recommendation.
Timeframe	The pattern's estimated resolution horizon — intraday, swing (3–10 days), or positional (weeks) — to align with the investor's own holding preference.

Confidence Score	A 0–100 composite score combining pattern quality, volume confirmation, market breadth alignment, and sector momentum. Higher scores indicate more reliable setups.
Visual Annotation	Interactive chart overlay with pattern boundaries, necklines, targets, and stop zones marked. Rendered natively on the ET Markets chart widget.

4.5 Sample Signal Card — Illustrative Example

SAMPLE SIGNAL CARD

Reliance Industries (RELIANCE) · NSE · Daily Chart

Pattern: Cup and Handle — Breakout Imminent

Confidence Score: 84 / 100 · Timeframe: Positional (3–6 weeks) · Detected: Today, 9:45 AM IST

What is happening?

Reliance Industries has formed a classic Cup and Handle pattern on the daily chart over the past 14 weeks. The stock fell from ₹2,980 to ₹2,710, recovered back to ₹2,975, and is now consolidating in a tight range between ₹2,940–₹2,980 — the handle phase. A sustained close above ₹2,985 with volume at least 1.5× the 20-day average would confirm the breakout.

What does history say?

This pattern has resolved bullishly 71% of the time on Reliance Industries across 23 historical occurrences over the past 5 years. Average upside on confirmed breakouts: 12.4% in 28 trading days.

Suggested Action Zone (not investment advice):

Entry on breakout: ₹2,985–₹3,010 | Stop Loss: ₹2,890 | Target 1: ₹3,220 | Target 2: ₹3,360

5. Technical Architecture

5.1 System Overview

The Chart Pattern Intelligence engine is architected as a real-time streaming microservice that sits between ET Markets' existing market data pipeline and the frontend client applications. It consists of four primary subsystems:

#	Subsystem	Responsibility
1	**Data Ingestion Layer**	Consumes real-time OHLCV tick data from NSE feed via WebSocket / FIX protocol. Normalises, validates, and writes to time-series store (InfluxDB/TimescaleDB). Handles corporate action adjustments automatically.

2	**Pattern Detection Engine**	Python-based computation engine using vectorised NumPy/Pandas operations and a proprietary rule engine. Runs full NSE universe scans every 5 minutes intraday; full overnight scan for positional patterns. Pattern results written to Redis for low-latency reads.
3	**AI Explanation Layer**	LLM-powered text generation (fine-tuned on Indian equity market corpus) that generates the plain-English signal narrative for each detected pattern. Output is cached per pattern instance. Hindi translation via NMT layer. Hallucination guardrails via structured prompt templates and numeric-value grounding.
4	**Back-test Analytics Store**	Pre-computed historical success rates for every pattern × stock combination stored in a columnar database (BigQuery / Redshift). Updated weekly. Powers the "X% success rate over Y occurrences" display without live computation overhead.

5.2 Data Requirements

- Historical OHLCV data: Minimum 5 years of daily data for all NSE-listed equities (~2,000 symbols).
- Intraday OHLCV: 1-minute and 5-minute granularity for the current trading session.
- Volume data: Delivery volume, total volume, and open interest for F&O-eligible stocks.
- Corporate actions: Dividend, split, bonus, rights — for accurate price adjustment.
- Index constituents: Nifty 50, Nifty 500, sectoral indices for relative strength calculations.

5.3 Performance Requirements

Metric	Target	Acceptable Minimum
Pattern detection latency (intraday)	**< 5 minutes from candle close**	< 15 minutes
Signal card page load (P95)	**< 800ms**	< 2,000ms
NSE universe scan coverage	**100% of NSE equities**	Nifty 500 minimum at launch
Back-test data depth	**5 years / stock**	3 years / stock
Push notification delivery	**< 30 seconds from detection**	< 2 minutes
System uptime (market hours)	**99.9%**	99.5%

6. UX & Product Experience Requirements

6.1 Entry Points

- Pattern Screener Hub: A dedicated section on ET Markets web and app where users can filter the full NSE universe by pattern type, timeframe, sector, and confidence score.
- Stock Page Integration: Every stock's detail page on ET Markets will display an active pattern card (if any pattern is currently detected) at the top of the Technical Analysis section.
- Watchlist Alerts: Users with watchlists receive a daily morning digest (7:30 AM IST) and real-time push notifications for high-confidence (80+) patterns on their watched stocks.
- Market Pulse Widget: A homepage widget surfacing the top 5 patterns of the day across all cap sizes.

6.2 Personalisation Requirements

- Users can set preferred timeframes (intraday / swing / positional) to filter the pattern feed to their trading style.
- Users can choose minimum confidence thresholds (e.g., show me only patterns scoring 70+).
- Language preference: English or Hindi for signal explanations.
- Notification preferences: push, email, or in-app only, with per-stock and per-pattern-type granularity.

6.3 Regulatory & Compliance Requirements

SEBI Compliance Note

All signal cards must carry a prominent disclaimer: "This is not investment advice. Pattern analysis is for educational and informational purposes only. Consult a SEBI-registered investment advisor before making investment decisions. Past pattern performance does not guarantee future results." The product must not use language that constitutes a buy, sell, or hold recommendation under SEBI's Investment Adviser Regulations, 2013.

7. Success Metrics & KPIs

7.1 Primary Product Metrics

Metric	3-Month Target	6-Month Target	12-Month Target
Weekly Active Users (Pattern Feature)	**100K**	500K	2M
Signal Cards Viewed per DAU	**3.0**	5.0	7.0
Push Notification CTR	**8%**	12%	15%

D7 Retention (Feature Cohort)	**35%**	42%	50%
NPS (Feature-Specific)	**40+**	50+	60+
Pattern Accuracy (Back-test δ ≤ 5%)	**80%**	85%	88%

7.2 Business Metrics

- Premium Conversion: Pattern Intelligence will be the primary driver of ET Markets Premium subscription conversion. Target: 15% of feature WAU converting to paid within 6 months.
- Session Depth: Users accessing pattern features should show a measurable increase in pages-per-session and time-on-site compared to the baseline cohort.
- Advertiser Value: High engagement and dwell time on pattern pages provides premium inventory for financial services advertisers.

8. Delivery Roadmap

Sprint / Week	Milestone	Deliverables
W1–W2	**Data Foundation**	Set up TimescaleDB, ingest 5-year NSE historical OHLCV, validate data quality, corporate action adjustments.
W3–W4	**Pattern Engine v1**	Implement 6 core pattern detectors (Cup & Handle, H&S, Double Top/Bottom, Golden/Death Cross, RSI Divergence). Unit test against labelled historical data.
W5–W6	**Back-test Store**	Run historical back-test for all 6 patterns across full NSE universe. Load results into analytics store. Build success-rate lookup API.
W7–W8	**AI Explanation Layer**	LLM prompt engineering, fine-tuning on Indian market corpus, guardrails implementation, Hindi translation pipeline. A/B test explanation quality.
W9–W10	**Frontend Integration**	Signal card UI component, chart annotation overlay, screener hub page, stock page integration. Mobile-responsive design. Accessibility review.
W11	**Notifications**	Push notification pipeline, user preference centre, morning digest email, in-app notification centre.
W12	**Beta Launch**	Closed beta with 10,000 ET Markets power users. Feedback loops, accuracy monitoring, performance tuning.

W13–W14	**GA Launch**	Staged rollout to full ET Markets user base. Marketing campaigns, PR, influencer partnerships. 24/7 ops monitoring live.

9. Risks & Mitigations

#	Risk	Severity	Mitigation
1	Regulatory misclassification as Investment Advice under SEBI IA Regulations	HIGH	Engage SEBI-registered legal counsel before launch. Implement strict disclaimer framework. Avoid first-person recommendation language in all AI-generated content. Add opt-in disclosure for users.
2	Low pattern accuracy damaging user trust	HIGH	Rigorous back-testing pipeline with labelled ground-truth data. Launch with only high-confidence patterns first. Pattern accuracy dashboard for internal monitoring. User feedback loop on every signal card.
3	LLM hallucination producing incorrect price levels or statistics	HIGH	All numeric values (price levels, success rates, dates) are injected from the structured data layer — never generated by the LLM. LLM is restricted to narrative template slots only.
4	NSE data feed disruptions causing stale signals	MED	Implement data freshness timestamps on every signal card. Automatic staleness warning if pattern detection is > 15 minutes behind. Fallback to previous close data with clear labelling.
5	Over-reliance by users leading to financial losses and reputational damage	MED	Educational content embedded alongside every signal explaining the probabilistic nature of technical analysis. First-use onboarding with risk education. Links to SEBI Investor Awareness resources.

10. Assumptions & Dependencies

10.1 Assumptions

- ET Markets has or can obtain a reliable NSE live feed with sub-minute OHLCV granularity at commercially viable cost.
- At least 5 years of adjusted historical OHLCV data is available for the full NSE equity universe.
- The ET Markets app infrastructure supports push notifications on iOS and Android.
- SEBI compliance review will be completed before the public launch of the pattern screener.

10.2 Dependencies

- NSE / Broker data vendor API for real-time and historical data.
- Cloud GPU instances for back-test computation and LLM inference.
- ET Markets charting library supporting annotation overlay API.
- Legal team review and SEBI counsel sign-off on disclaimer language.
- Existing ET Markets authentication and user preference infrastructure.

11. Open Questions & Decisions Required

#	Question	Owner	Due
1	Should Chart Pattern Intelligence be a free feature for all users, or a premium-only feature?	Product + Business	W1
2	Which LLM provider (OpenAI, Anthropic, Google, or self-hosted) best balances cost, latency, and compliance for the explanation engine?	Engineering Lead	W2
3	What is the acceptable false positive rate for pattern detection before users lose trust?	PM + Data Science	W3
4	Should the suggested action zone (entry/stop/target) be shown to all users, or only after a risk-tolerance questionnaire?	Legal + Product	W1
5	Should pattern alerts be enabled by default (opt-out) or require user opt-in to comply with TRAI/telecom regulations on commercial communications?	Legal + Engineering	W2

Appendix — Glossary of Terms

Term	Definition
OHLCV	Open, High, Low, Close, Volume — the five data points recorded for each time period of a security's price history.
NSE	National Stock Exchange of India — the largest stock exchange in India by daily trading volume.

Demat Account	Dematerialised account — a digital account that holds shares and securities in electronic form.
RSI	Relative Strength Index — a momentum oscillator measuring the speed and change of price movements on a 0–100 scale.
MACD	Moving Average Convergence Divergence — a trend-following momentum indicator showing the relationship between two moving averages.
ATR	Average True Range — a measure of market volatility used to size stop-loss distances and position sizes.
SIP	Systematic Investment Plan — a method of investing a fixed amount regularly in mutual funds.
F&O	Futures and Options — exchange-traded derivative instruments available on select NSE equities and indices.
Back-test	The process of testing a trading strategy or pattern detection rule against historical data to estimate its performance.
SEBI IA Regs	Securities and Exchange Board of India (Investment Advisers) Regulations, 2013 — the regulatory framework governing who can provide investment advice in India.

— End of Document —

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